Name: _____

Astronomy 3

Date:

Time:

Total marks available: 24

Total marks achieved: _____

Questions

Q1.

Scientists no longer accept the geocentric model of the universe but it was the accepted theory for hundreds of years.

Explain why the evidence available at the time supported the geocentric model.

(3)

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(Total for question = 3 marks)

Q2.

* Figure 14 shows two ice skaters during a sequence in their performance.

Figure 14

The man stays at the same place on the ice throughout the sequence.

At the start of the sequence, the woman is moving at a constant speed around the man while the man holds her arm.

After she has gone round the man several times, the man lets go of the woman's arm.
The sequence ends a few seconds later.

Explain the motion of the woman, in terms of the forces acting and the effects on her motion, for the whole sequence.

(6)

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(Total for question = 6 marks)

Q3.

Observations of the Universe can be made using telescopes on Earth.

Explain why some telescopes are located on satellites that orbit the Earth.

(2)

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(Total for question = 2 marks)

Q4.

* The Sun is likely to remain as a main sequence star for 2000 million years.

Explain why the Sun does not explode now.

(6)

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(Total for question = 6 marks)

Q5.

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

Explain why Vesta is accelerating even when it is travelling at a constant speed.

(2)

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(Total for question = 2 marks)

Q6.

Which row of the table is correct for both force and velocity?

(1)

	force	velocity
☐ A	scalar	scalar
☐ B	scalar	vector
☐ C	vector	scalar
☐ D	vector	vector

(Total for question = 1 mark)

Q7.

Figure 6 shows a satellite orbiting the Earth.

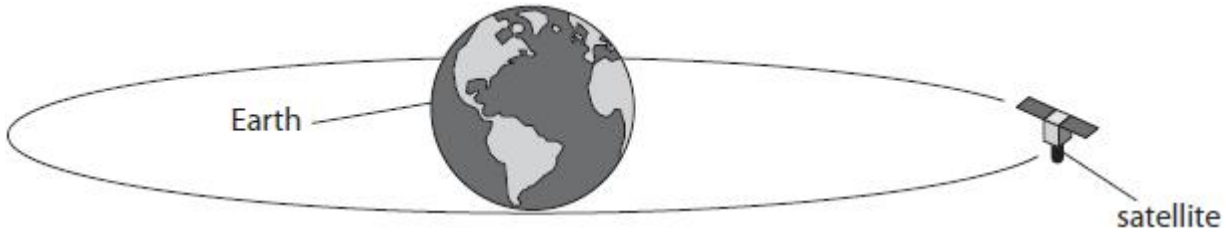

Figure 6

(i) State the name of the force that keeps the satellite in orbit around the Earth.

(1)

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(ii) Draw an arrow on Figure 6 to show the direction of the force acting on the satellite, that keeps the satellite in orbit around the Earth.

Label this arrow 'F'.

(1)

(Total for question = 2 marks)

Q8.

This picture shows Galileo holding a telescope.

(a) Explain the purpose of the eyepiece in a telescope.

(2)

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(b) Galileo drew pictures of his observations of Jupiter.
Nowadays we can take photographs.
Suggest how photographs would have helped Galileo.

(1)

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(c) The telescope collects light reflected from Jupiter.
The light has a frequency of 4.30×10^{14} Hz and a speed of 3.00×10^{8} m/s.
Calculate the wavelength of the light.

(3)

wavelength =..... m

*(d) Galileo's observations of the moons of Jupiter disproved the geocentric model.
However, these observations were not enough to prove the heliocentric model of the Solar System.
Explain why Galileo's observations disproved one model but were not enough to prove the other model.

(6)

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(Total for Question = 12 marks)

Q9.

State **two** advantages of using a camera, rather than the unaided eye, for studying stars.

(2)

1

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2

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Q10.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

Vesta orbits the Sun between the orbits of

(1)

☐ **A** Venus and Earth

☐ **B** Earth and Mars

☐ **C** Mars and Jupiter

☐ **D** Jupiter and Saturn

(Total for question = 1 mark)

Q11.

(i) A long time ago, scientists believed that the Earth was at the centre of the Solar System.

Evidence has since proved that the Sun is at the centre of the Solar System.
State **one other** idea about the Solar System that **has** changed over time.

(1)

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(ii) Figure 4 shows data for some of the planets of the Solar System.

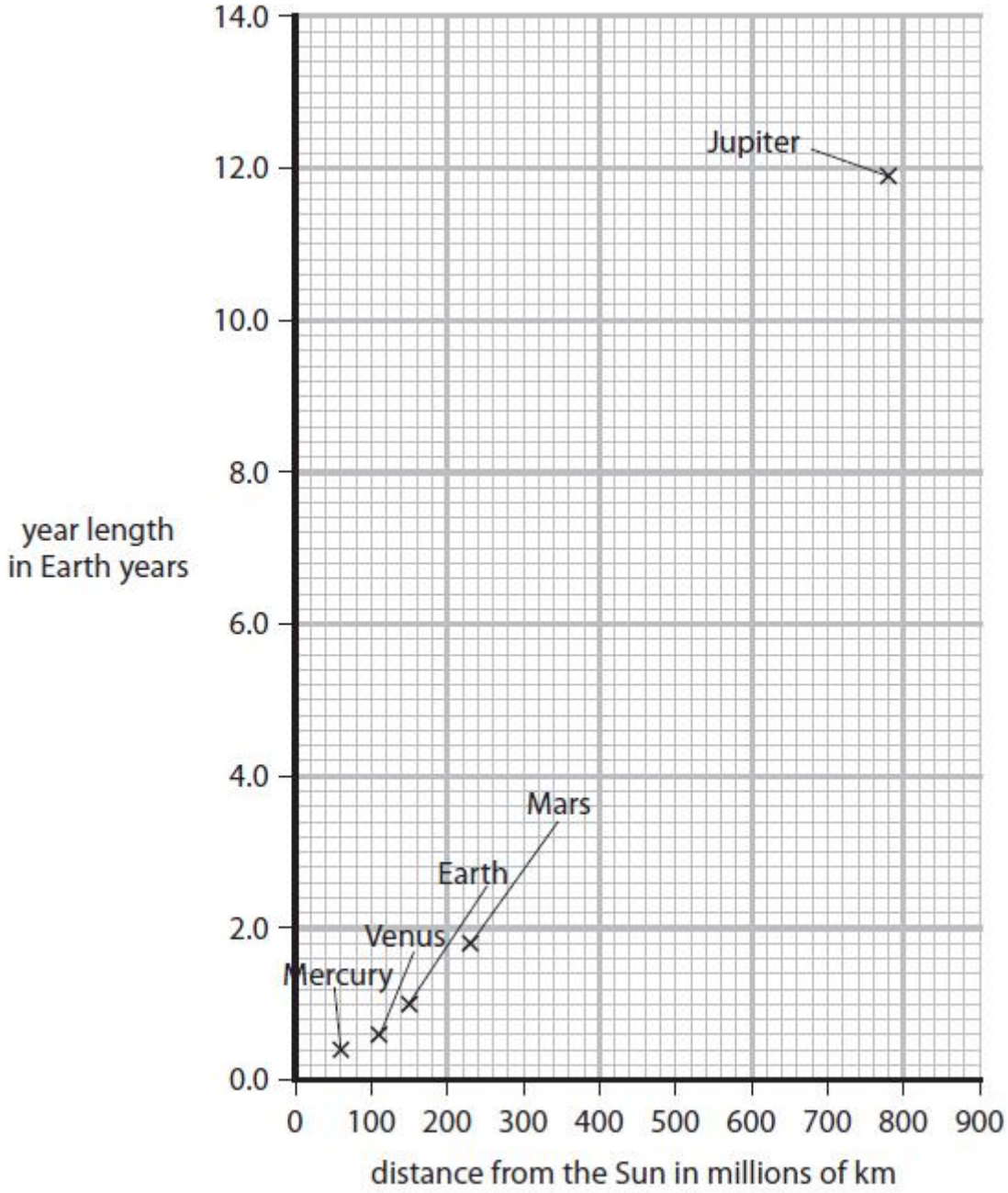

Figure 4

Ceres is an asteroid that orbits the Sun between Mars and Jupiter.
It takes Ceres 4.6 Earth years to make one orbit of the Sun.
Use the graph to estimate the distance of Ceres from the Sun.
Show your working.

(3)

distance of Ceres from the Sun = ... millions of km

(Total for question = 4 marks)